

Evaluating pharmacist interventions in a health-system specialty pharmacy: impact of embedded clinic pharmacists on specialty pharmacist intervention patterns

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Background

- Pharmacist involvement in patient care within health-system specialty pharmacies (HSSPs) has been shown to enhance patient outcomes and reduce costs.¹
- Capturing pharmacist interventions is common², but data is lacking for utilization of this metric for benchmarking, evaluating HSSP pharmacist workload, and leveraging for staff expansion.
- Although several sources offer guidance on establishing HSSP metrics and clinical benchmarks, they may not be applicable across different staffing models.³ West Virginia University Hospitals Specialty Pharmacy (WVUHSP) operates as a fully centralized model, while our broader health system includes pharmacists in select clinic settings. Available publications do not address variability in clinic settings, such as embedded pharmacists, that may affect HSSP pharmacist intervention frequency.
- Lastly, the workload impact of HSSP interventions is unclear, necessitating proper accounting for complexity and time spent on these activities.

Objective

Primary: Compare the proportion of patients receiving HSSP pharmacist interventions across oncology clinic types characterized by having:



Methods

- Setting:** Oncology clinics serviced by WVUHSP
- Timeframe:** January 1st, 2025 through March 31st, 2025
- Design:** Retrospective chart review of interventions documented by pharmacists in patient management software
 - Inclusion criteria:** All patients with one fill of specialty medication in the category of oncology during study timeframe
 - Exclusion criteria:** Patients who did not receive a fill with WVUHSP or did not receive a specialty medication for oncology

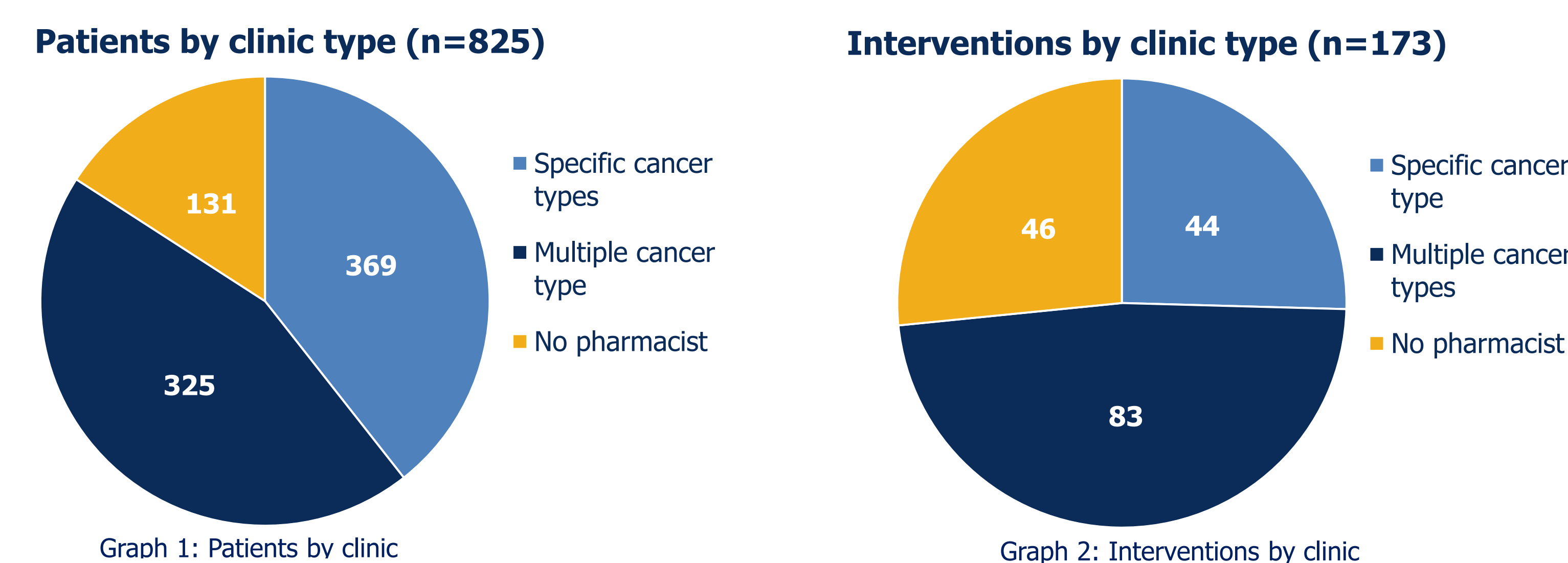
Results

Patient demographics (n=825)	
Female % (n)	60.7% (501)
Age in years (median), (range)	56 (3->89)

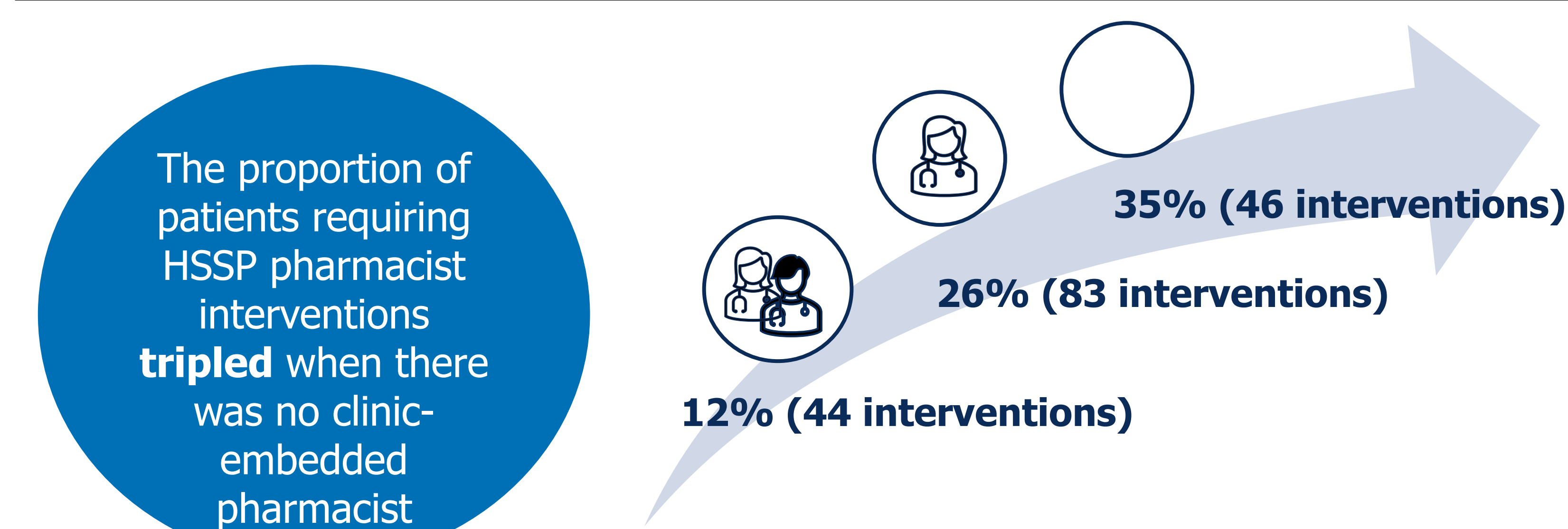
Table 1: Patients receiving at least one fill of specialty medication in oncology

Clinic type	Patients	Pharmacist interventions
Specific cancer types	369	44
Multiple cancer types	325	83
No pharmacist	131	46
	825	173

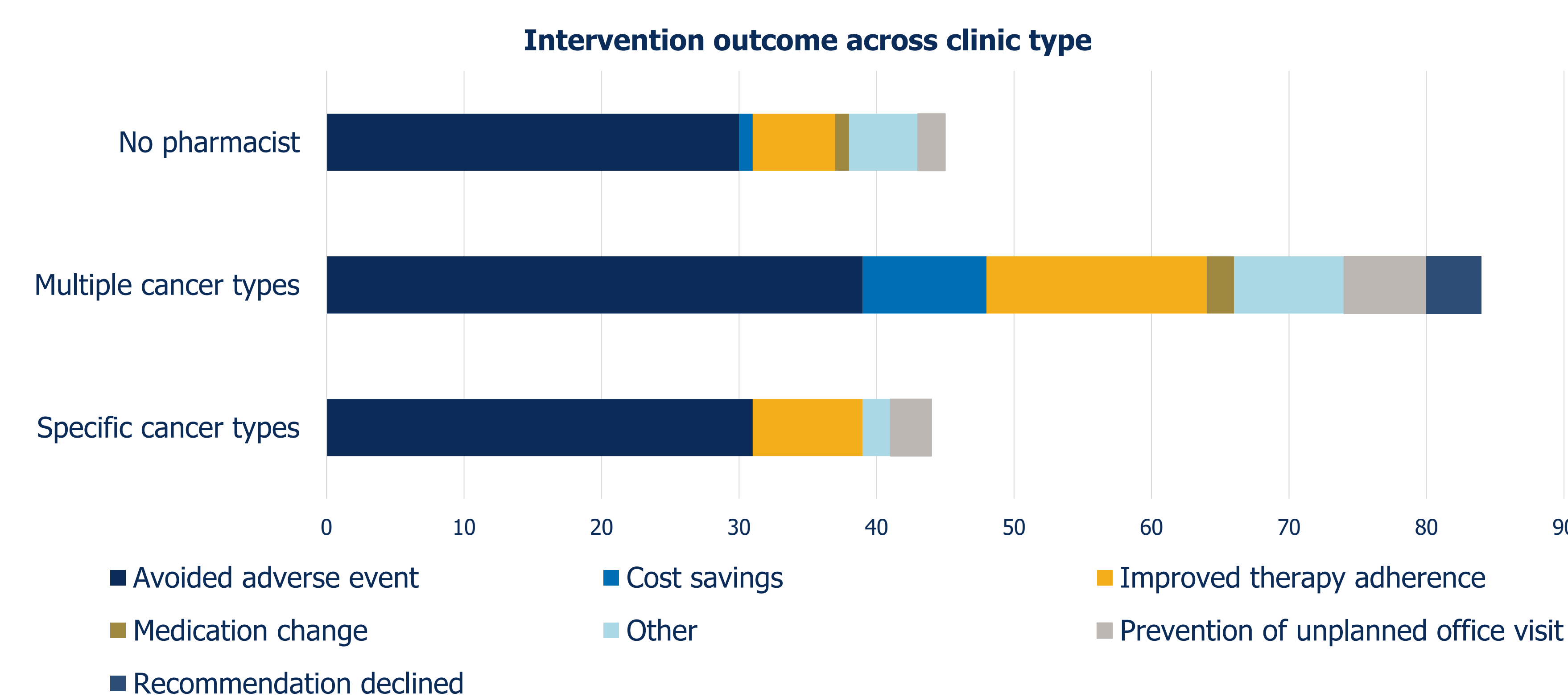
Table 2: Patients receiving one fill of specialty medication for oncology with one or more HSSP pharmacist intervention within the study timeframe



Interventions by clinic type (n=173)



Graph 3: Patients receiving one fill of specialty medication for oncology with one or more HSSP pharmacist intervention by clinic type



Graph 4: Intervention outcomes by clinic type

Outcome of intervention (n=173)	
Avoided adverse event	100
Cost savings	10
Improved therapy adherence	30
Medication change	3
Other	15
Prevention of unplanned office visit or hospitalization	11
Recommendation declined	4
Total	173

Table 3: Patients receiving one fill of specialty medication for oncology with one or more HSSP pharmacist intervention within the study timeframe

97.7% of pharmacist recommendations were accepted

Conclusion

- Clinics without embedded pharmacists had the highest percentage of HSSP pharmacist interventions (35%), suggesting a potential gap in care coordination and reinforcing the critical value of HSSP pharmacist.
- Utilizing the number of prescriptions filled by a HSSP does not fully capture pharmacist workload. Workload associated with initial and follow-up care plans is more structured, while pharmacist interventions are more variable. This ratio of interventions per patient may be a potential proxy to reflect the HSSP pharmacist involvement across differing clinic settings.
- HSSPs face increasing uncertainty due to challenges within the 340B program, evolving legislation, and diminishing payer margins—factors that complicate appropriate staffing and threaten the sustainability of high-quality patient care. Quantifying intervention workload relative to prescription volume may provide a more robust framework for predicting staffing needs.
- This three-month study offers preliminary insight but highlights the need for longer-term analysis.

Future directions

Evaluate intervention outcomes across clinic types in other disease areas

Evaluate time spent and complexity of interventions across clinic types

Incorporate cost analysis to better justify staffing models and support strategic growth

References

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